BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021 • Paper Code: DIK-SSC-IX-2021(A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 12 | **Paper Code:** DIK-SSC-IX-2021(A)

Q1. Choose the correct answer from four possible answers (A,B,C,D) and fill the correct answer on MCQ's sheet separately.

Q1. The study of insects is called _____

(A) Entomology
(B) Parasitology
(C) Embryology
(D) Morphology

Q2. _____ is the basic and smallest unit of classification.

(A) Species
(B) Genus
(C) Family
(D) Order

Q3. In 1665 _____ was the first person to observe cell with a small microscope.

(A) Robert Brown
(B) Robert Haeckel
(C) Robert Hooke
(D) Robert Darwin

Q4. In chloroplast, the stack of thylakoids is known as _____

(A) Stomata
(B) Chlorophyll
(C) Stroma
(D) Granum

Q5. Division of cytoplasm is known as _____

(A) Cytokinesis
(B) Karyokinesis
(C) Cytogenesis
(D) Karyogenesis

Q6. Exchange of parts between non-sister chromatids of homologous chromosomes is called _____

(A) Chiasmata formation
(B) Synapsis
(C) Disjunction
(D) Crossing over

Q7. Some enzymes require non-protein molecules called_____

(A) Coenzymes
(B) Cofactors
(C) Inhibitors
(D) Substrate

Q8. Light reaction takes place in the _____ of chloroplast.

(A) Granum
(B) Stroma
(C) Stomata
(D) Lenticies

Q9. During glycolysis, glucose is broken down into two molecules of_____

(A) Lactic acid
(B) Pyruvic acid
(C) Amino acid
(D) Formic acid

Q10. Vitamin C is also known as _____

(A) Retinol
(B) Glutamic acid
(C) Ascorbic acid
(D) Formic acid

Q11. The inner boundary of cortex is known as_____

(A) Ectoderm
(B) Endoderm
(C) Endodermis
(D) Epidermis

Q12. A person with blood group A contains antigen_____

(A) A only
(B) B only
(C) AB both
(D) O (Nil)

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 21 | **Paper Code:** DIK-SSC-IX-2021(A)

Q2. Attempt any 8 questions of the following:

(i) Briefly describe multicellular organization as an example of cellular organization.

(ii) What is Biological method? Name the steps involved in this method.

(iii) Give any four difference between prokaryotes and eukaryotes.

(iv) What are meristematic tissues? Briefly describe their types.

(v) What is cell cycle? Name its main stages.

(vi) Briefly describe induced fit model of enzyme action.

(vii) Briefly describe ATP as the energy currency of cell.

(viii) What is aerobic respiration? Name its three main steps.

(ix) What is meant by mineral nutrition? Differentiate between macronutrients and micronutrients.

(x) What is transpiration? Why transpiration is considered as a necessary evil?

(xi) What is blood? Briefly describe the composition of the plasma of blood.

SECTION – C

Maximum Marks: 24

> Note: Attempt any 3 questions: Each carry 8 marks.

Q3.

(a) Briefly describe conclusion and reporting of results as s step of biological method.

(b) What are phloem tissues? Differentiate between sieve tube and companion cell. Also draw their diagrams.

Q4.

(a) Draw a table showing the distinguishing characteristics of kingdom Monera, Protista and Fungi.

(b) Briefly describe lactic acid fermentation.

Q5.

(a) Briefly describe intracellular and extra cellular enzymes.

(b) Briefly describe the significance of mitosis.

Q6.

(a) Write short note in vitamin-D.

(b) Describe and draw the internal structure of human heart.